

# Machine Learning

Non-parametric Regression: k-NN Method and its variants. Bias-Variance trade-off for k-NN Regression. Mean (Absolute) Test Error.

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ML Research



## ① Non-parametric Regression

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# Non-parametric Regression

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## Assumption

Close objects correspond to similar answers

# Non-parametric Regression

The simplest model

We approximate the desired dependence by a constant in some neighborhood

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# Non-parametric Regression

## The simplest model

We approximate the desired dependence by a constant in some neighborhood

## Nadaraya-Watson kernel regression<sup>1</sup>

If there are several objects from the training sample in the vicinity of the point, then it is reasonable to **use the weighted average** as a prediction of the algorithm

$$a(x) = \frac{\sum_i y_i \omega_i(x)}{\sum_i \omega_i(x)},$$

where  $\omega_i(x) = K_h(x, x_i)$ , a function  $K_h$  is called a **kernel** with smoothing window width  $h$ .

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## k-NN Regression: simplest prediction method

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## k-NN Regression prediction

Let us have for every  $x_0$   $k$  nearest neighbors  $(x_1, \dots, x_k)$  with the ground truth labels  $(y_1, \dots, y_k)$ . Then the Nadaraya-Watson kernel regression formula will transform into the following:

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**Note:** It means we are just averaging the labels of  $k$  nearest neighbors.

## Examples of more complicated Kernels

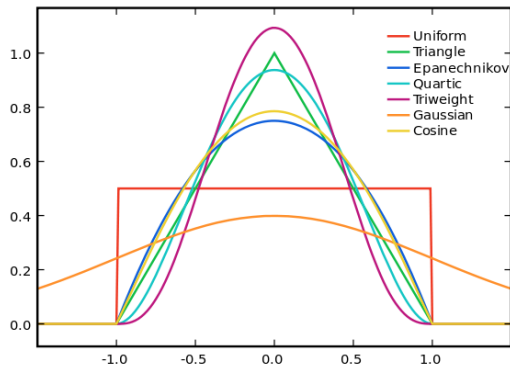
- $K_h(x, x_i) = K\left(\frac{\|x - x_i\|}{h}\right)$

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## Examples of more complicated **Kernels**

- $K_h(x, x_i) = K\left(\frac{\|x - x_i\|}{h}\right)$
- Typical Examples: <sup>2</sup>



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## k-NN Regression: Mean Test Error

- Suppose that we have  $L$  test points  $(x_1^t, \dots, x_L^t)$  with the corresponding ground truth labels  $(y_1^t, \dots, y_L^t)$

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**Note:** It means we are just averaging the absolute error for every point-wise prediction across the test set.

## Reminder: bias-variance tradeoff

### Definitions

Let  $y = y(x) = f(x) + \varepsilon$  be the target dependence, where  $f(x)$  is the deterministic function,  $\varepsilon \sim N(0, \sigma^2)$  and  $a(x)$  is the machine learning algorithm.

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$$E(y - a)^2 = \sigma^2 + \text{variance}(a) + \text{bias}^2(f, a)$$

# Bias and Variance of k-NN Regression

## Bias

$$\text{bias}^2(f, a) = (E(f(x_0) - a(x_0)))^2 = \left( f(x_0) - \frac{1}{k} \sum_{i=1}^k f(x_i) \right)^2$$

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## Variance

$$\begin{aligned} \text{Variance}(a) &= D \left( \frac{1}{k} \sum_{i=1}^k y(x_i) \right) = \frac{1}{k^2} D \left( \sum_{i=1}^k y(x_i) \right) = \\ &= \frac{1}{k^2} D \left( \sum_{i=1}^k (f(x_i) + \varepsilon_i) \right) = \frac{1}{k^2} D \left( \sum_{i=1}^k f(x_i) \right) + \frac{1}{k^2} D \left( \sum_{i=1}^k \varepsilon_i \right) = \\ &= 0 + \frac{1}{k^2} k \sigma^2 = \frac{\sigma^2}{k} \end{aligned}$$



## Bias-Variance tradeoff of k-NN Regression

$$Error(x_0) = E(a(x_0) - y(x_0))^2 = \left( f(x_0) - \frac{1}{k} \sum_{i=1}^k f(x_i) \right)^2 + \frac{\sigma^2}{k} + \sigma^2$$

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**Note:** Under “reasonable assumptions” the bias of the 1-NN estimator vanishes entirely as the size of the training set approaches infinity

# Mandatory external links to read

- 1 Read the Introduction to K-Nearest Neighbor (kNN) algorithm for the Regression task
  - ▶ Main source, more thorough explanation, and rigorous algorithm overview.
  - ▶ A video covering a practical example.

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- To compute the k-NN Regression Prediction we are averaging the nearest neighbors labels,
- To compute the k-NN Regression Mean (Absolute) Test Error we are averaging the absolute error for every point-wise prediction across the test set.

# Thank you!